THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONAE PRE MOCK JOINT EXAMINATION
FORM FOUR- CHEMISTRY

TIME: 3 HOURS **May, 2021**

INSTRUCTIONS:

1. This paper consists of sections A, B and C

2. Answer all questions in section A and B and one question in section C

3. Non-programmable calculators are allowed in examination room

4. Write your examination number on your answer sheet(s)

5. The following constants may be used;

.Atomic masses

H=1, C= 12, O=16, S=32, Cl=35.5, Ca= 40, Cu= 64

.Avogadro's constant=6.02×10^{23} particles

.GMV at s.t.p =$22.4dm^3$

. 1 Faraday = 96500C

.Specific heat capacity of water, Cw=4.18KJ/Kg/Kg

.Density of pure water,=$1g/cm^3$ or $1000kg/m^3$

SECTION A (15 MARKS)

1.For each of the following items (i)-(x) below choose the letter of the most correct answer from among the alternatives given and write its letter in the answer sheet provided.

(i)During volumetric analysis the solution added to the burette is referred to as the
A. Acid and base solution
B. Analyte
C. secondary volume
D. titrant
E. soluble solution

(ii)Which of the following reactions, $Cu^{2+}_{(aq)} + Zn_{(s)} \longrightarrow Zn^{2+}_{(aq)} + Cu_{(s)}$ represents?

A. Neutralization
B. Decomposition
C. Precipitation
D. Synthesis
E. Displacement

(iii)An example of homologous series is
A. ethene, ethyne and propyne
B. propane, butane and pentyene
C. ethane, propane and butyne
D. ethane, propane and butane
E. methane, ethane and propane

(iv) The volume of 0.2M H_2SO_4 acid required to neutralize completely 25.00cm^3 of 0.05M KOH is
A.0.626cm^3
B.12.500cm^3
C.6.315cm^3
D.3.125cm^3
E.6.125cm^3

(v)A wasp sting is alkaline. The solution to help easily the pain by neutralizing the alkali would one of with PH of;
A.5
B.8
C.7
D.10
E.13

(vi) Musa was doing investigation but unfortunately he found the results, obtained do not support his hypothesis. What advice could you give?

A. Give ideas for further testing to find a solution
B. leave out the results
C. change his experiment
D. identify a new problem
E. draw conclusion

(vii)Which of the following statements is correct about the water gas?

A.it is poisonous
B. contains hydrogen
C.is the same as biogas
D. contains hydrogen and nitrogen
E. contains carbon monoxide and hydrogen

(viii)The only metal which does not react with diluted hydrochloric acid is.

A. magnesium
B. aluminum
C. copper
D. zinc
E. sodium

(ix)An element in the periodic table with atomic number 18 belongs to which of the following?

A. group I and period I
B. group 0 and period III
C. group III and period III
D. group V and period IV
E. group VII and period IV

(x)Which of the following is a covalent compound?

A. water
B. sodium hydroxide
C. aluminum oxide
D. calcium fluoride
E. zinc nitrate

2. Match the items in list A with the correct responses in list B by writing the letter of the correct responses besides the item number in the answer sheets provided

LIST A		LIST B
I	Is the application of calcium in various forms of soil in order to increase the PH of the soil	A. Green manure
II	Contains two more of the primary macro nutrients	B. Farm yard manure
III	Trace elements	C. Straight fertilizers
IV	The capacity of the plant to support plant growth	D. Soil fertility
V	The nutrients content of manures varies from one animal to another	E. Micro- nutrients
		F. Compound fertilizers
		G. Soil productivity
		H. Liming
		I. Leaching
		J. Macro nutrients

SECTION B (70MARKS)

Answer all questions in this section

3. (a) (i)Name the element present in pure charcoal

(ii)Explain why it is dangerous to use charcoal stove in poorly ventilated room?

(b) (I) describe what happen to it after sometimes?

(ii) Write a balance chemical equation to show how bleaching power is prepared

4. (a) Apart from iron ore (haematite),name two other raw materials fed into the blast furnance in the extraction of iron metal

(b) Write the equation to show how the ore is reduced in the blast furnance

(c)When sodium nitrate is heated, it produces sodium nitrite and gas Y.

(i)Identify gas Y

(ii)Name the type of the reaction undergone by sodium nitrate

5(a) a concentrated solution of sodium chloride was electrolyzed using platinum electrodes

(i) State what is observed from at the anode

(ii)Explain your observation in (i) above

(b)Butane is a saturated hydrocarbon

(i)what does the term saturated hydrocarbon mean?

(ii)Give the equation for complete combustion of butane when burnt in plentiful supply of air

6. (a)State Avogadro's law

(b)A litre of methane cost Tsh 60/.Calculate the cost of methane required to produce 2000KJ of heat.(1mole of a gas occupy 24dm^3 at room temperature, the heat of combustion of methane is -882KJ/MOL)

7. (a) Write an equation to show how ethanol can be prepared from glucose

(b) State how a sample of ethanol obtained from the product of the reaction in (a) can be purified

(c) Write an equation for the complete combustion of ethanol

8. (a) Briefly explain four methods of preventing rusting

(b) The body of a ship is painted to prevent rusting.

(i) how does painting stop rusting?

(ii) block of zinc are attached to the hull of a ship to preventing rusting. Explain

(iii) Why might a ship in a sea water rust faster than one that is only used in a fresh water lake?

9. (a)Name three factors that affect the rate of a chemical reaction

(b)How does the increase in;

(i) temperature

(ii) pressure

Affect the yield of ammonia in the following reaction?

$N_{2(g)}+3H_{2(g)} \rightleftharpoons 2NH_{3\,(g)}$ Heat change = -92kj/Mol^{-1}

(c) Name the catalyst used in the above reaction

10. (a)In terms of electrons, define the following terms

(i) Oxidation

(ii) Reduction

(b)Calculate the oxidation number of the underlined elements in the following compounds

(i)$Na_2\underline{Cr}O4$

(ii)$\underline{N}O_3^-$

(c)During electrolysis of copper sulphate solution, a current of 0.5A was passed for 3860s.The amount of copper deposited was 0.64g

(i) calculate the charge carried by the copper ion

(ii) write an ionic equation for the reaction

11. (a)Differentiate between;

(i) Solution and suspension

(ii) isobars and isotopes

(iii) isotones and allotropes

(b)Complete and balance the following chemical reactions

(i)$Mg_{(s)} + N_{2\,(g)} \longrightarrow$

(ii)$NH_4NO_{3(S)} \longrightarrow$

(iii)$NO_{\,(g)} + H_{2(g)} \longrightarrow$

(c) outline three uses of hydrogen gas

12. (a)Define the following terms;

(i) empirical formula

(ii) molecular formula

(iii) general formula

(b)Calculate the molarity of calcium ions, and chlorine ions in a $CaCl_2$ solution

(c)246g of a hydrated salt with formula of $MSO_4.nH_2O$,on heating produced 120g of anhydrous MSO_4.Calculate the number of moles of water of crystallization

SECTION C (15 MARKS)

Answer only one question from this section

13. (a) Why a firewood fuel is said to be better than gaseous fuel? Give four points

(b)Define;

(i) ignition point of a fuel

(ii) energy value of a fuel

(c)A mass of 40.00kg of petrol fuel was burnt in air; the heat produce was to heat 5 litres of water.Given that the heat value of petrol is 43640KJ/KG.what was the temperature change in water?

14. Describe three (3) causes of terrestrial pollution and four (4) methods of preventing terrestrial pollution.